Practical SOH Estimation Based on Feasible Feature Weighted Li-ion Battery Model

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***Abstract*—State-of-health (SOH) is one of the most important state that reflects the battery health status, which plays an important role in guaranteeing normal operation of cells in battery management system (BMS). Complicated aging mechanisms and working conditions make it difficult for accurate and efficient SOH estimation. In order to realize real-time accurate SOH estimation in actual application, this paper proposes a SOH-related health feature (HF) -weighted method for SOH estimation. The proposed SOH estimation method first extracts HFs from some easy-to-access data in actual application. These features comprehensively reflect the battery aging mechanisms. Then, a weighted SOH estimation model is developed from the extracted HFs of the battery. With the HFs extracted from the battery data, SOH can be estimated by the weighted SOH estimation model. Accelerated battery aging experiments are conducted to validate the proposed SOH estimation method. Experimental results show the effectiveness of the proposed method in accurately estimating the battery SOH.**

***Index Terms*—Battery, State-of-health (SOH), Health feature (HF), Weighted model, Battery aging**

I. Introduction

Compared to traditional fossil fuels, new energy sources are of great significance to achieve energy sustainability, environment protection, and economic growth. Lithium-ion batteries (LIBs), with their high energy density, long cycle life, and decent charge and discharge performance, are important clean power sources of new energy [1]–[3]. In recent years, the application of LIBs in electric vehicles, laptops, ships, aerospace, and other fields have shown a rapid growth trend [4]. For example, US government proposed 2 trillion strategic plan for clean energy projects calling for end to power plant emissions by 2035. However, LIBs may experience serious safety issues under abnormal operating conditions, such as over-charging, over-heating, accelerated aging, etc. These safety issues may lead to serious consequences such as thermal runaway, explosion, etc., and eventually cause significant impacts or even disasters to the systems, life and property, etc [5]. Therefore, it is crucial to develop a method that can conveniently, safely, reliably, and accurately estimate and predict the state-of-health (SOH) of LIBs.

SOH is an indicator that describes the health condition of a battery. It reflects the maximum volume of energy storage and deliverable capacity of the battery [6]. With an increasing demand for LIBs, SOH estimation plays an important role in battery longevity improvement, cost savings, system functionality maintenance, and safety operation, etc. SOH is an internal state of LIBs which cannot be directly measured and, therefore, needs a SOH estimation method. However, the accurate and practical estimation of SOH faces lots of challenges, especially in actual application, as it requires considering the comprehensive effects of factors such as charge and discharge cycles, temperature, operating conditions, battery design, etc [7]. Additionally, the cells are in form of pack to meet voltage and energy density requirement, which makes it inconvenient to get effective data to estimate the SOH of battery [8]. Therefore, there is a need to develop practical and accurate method for SOH estimation.

In recent years, different methods for SOH estimation have been developed and each of them has its own advantages and disadvantages [7], [8]. The SOH estimation methods mainly include model-based and data-driven methods [9]–[11]. Model-based methods refer to SOH estimation of the battery with physical model and semi-physical model. Least square-based methods and their improved derivatives offer simple implementation for model parameter identification and high accuracy for model calibration. The model-based methods usually offer decent performance in terms of accuracy and efficiency for different kinds of batteries in different working conditions. However, physical or semi-physical SOH models with high fidelity are usually highly nonlinear because they are described by many ordinary and partial mathematical equation [12], [13]. Therefore, a complex battery system always desires more intricate state estimation algorithms, and algorithm performance may be influenced by noise and uncertainty [14]. Although state estimation algorithms such

as Kalman filters (KFs) [15] and their variants have been widely employed to achieve real-time SOH, it is challenging and computing expensive to deploy efficient state estimation algorithms to these high-fidelity models.

With large datasets of battery, data-driven methods, including machine learning and deep learning methods, have shown great potentials to learn the battery aging mechanisms with impacts from the usage patterns and working conditions, such as depth-of-discharge (DoD), ambient temperatures and humanities, etc. With these advantages, some of the developed data-driven models can achieve promising SOH estimation performance with enough high-quality testing data [11], which reflect battery aging comprehensively. However, the quantity and quality of the training data significantly affect the model and estimation [16]. Besides, in actual application, it is not easy or even impossible to obtain enough high-quality data for the training. Moreover, the characteristics of the machine learning based model, is often referred as a 'black-box-model', may affect the model interpretability and transferability [17]. Additionally, researchers have proposed a more efficient SOH estimator that extracts the knee points of voltage transients in short-term current pulse tests as degradation features. With the optimization by the non-dominated sorting genetic algorithm II (NSGA-II) [18], a series of non-dominated solutions are obtained, and achieve good performance in terms of efficiency and accuracy of SOH estimation. It is worth noting that some of these methods make a trade-off between experimental resources occupation for the selection of effective health features (HFs) and SOH estimation performance. In general, more comprehensive experiments under different working conditions theoretically can generates enough high-quality data to establish a good SOH estimation model.

In this research, we propose a novel practical method for SOH estimation. First, easy-to-access data in actual application are selected for HFs extraction. Second, the HFs that comprehensively reflect battery aging mechanisms are extracted. Third, a weighted SOH estimation model that incorporates different HFs are developed. Finally, accelerated aging experiment data are used to validate the proposed model. Experimental results show promising performance of the proposed SOH estimation method. Our research aims to investigate the insights of easy-to-access HFs and find a practical way to develop a SOH estimation method for real applications.

The rest of the paper is organized as follows: Section II describes HFs extraction for SOH estimation. This includes experimental data analysis and HFs extraction. Section III presents the correlation between battery aging and HFs for model establishment, and the developed model validation. Section IV presents the conclusion.

II. Health features extraction

This section analyzes the internal electrochemical reaction process of LIBs to determine HFs for SOH with the consideration of accessibility and effectiveness of HFs. The following subsections will focus on experimental data analysis, HFs selection and extraction.

The first part presents experimental setup, battery accelerated aging experiment procedure design, and battery aging data analysis. The characteristics of the terminal voltage changing along with battery aging are analyzed to extract effective HFs.

SOH represents the current health state of the battery, which is usually defined as:

$$SOH_i = \frac{Q_i}{Q_0} \times 100\% \tag{1}$$

where i is the aging cycle index, SOH_i is SOH value at the i-th aging cycle, Q_0 is initial capacity of battery without cyclic aging, and Q_i is capacity at the i-th aging cycle. It is worth noting that the capacity Q is calculated at 1/3 C-rate referenced to the rated capacity under room temperature.

A. Experiment data analysis

To get the battery aging data for SOH model development, accelerated aging experiment are performed with Sanyo 14500 LiCoO2/graphite LIB samples with a rated capacity of 780 mAh. Fig. 1 shows the Arbin BT2000 test system used for accelerated aging experiment [6], which automatically samples the data of current and voltage at the sampling rate of 1 Hz. Table I summarizes the battery specification [6].

Fig. 1: Experimental setup: Arbin BT2000 test system

TABLE I: Battery specification

Specifications	Value
Height	$50.0 + 0.2$ mm
Diameter	$14.0 + 0.2$ mm
Weight	Approx.: 50 g
Rated capacity	780 mA h
Cutoff voltage of charging	4.2 V
Positive material	$LiCoO_2$
Negative material	Graphite

The accelerated battery aging experiment is performed by an Arbin BT2000 test system. The experiment test procedure is shown in Fig. 2 followed by the detailed steps:

Step1: Set the ambient temperature as room temperature and rest the battery for 1 hour to let it reaches thermal equilibrium;

Step2: Charge the battery with constant-current at 1 C-rate until the terminal voltage reaches 4.2 V;

Step3: Charge the battery with constant-voltage at 4.2 V until the charging current decreases to the cut-off value of 0.02 C-rate;

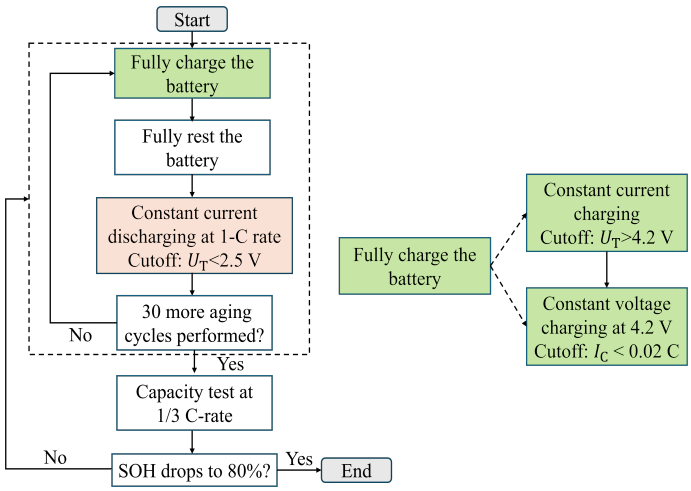

Fig. 2: Test procedure for the accelerated battery aging experiment. Note that U_{T} is the terminal voltage, and I_{C} is the cut-off current in the test procedure.

Step4: Rest for 1 hour to reach thermal equilibrium;

Step5: Discharge the battery at the 1 C-rate until the terminal voltage reaches the cut-off voltage of 2.5 V.

Step6: Repeats steps 2 to 5 for 30 aging cycles;

Step7: Fully charge the battery by steps 2 to 3 and rest it for 1 hour to obtain battery thermal equilibrium;

Step8: Discharge the battery at 1/3 C-rate, and calculate the capacity noted as Q_i, where index i is aging cycle number;

Step9: Repeat steps 1 to 8 until the capacity retention of Q_i drops to 80%;

Step10: Terminate the accelerated aging experiment if the capacity retention of Q_i drops to 80%.

The aging mechanisms are complicated and the working conditions of battery are unpredictable, which make it difficult to estimate battery capacity accurately. The battery capacity gradually decreases with the aging cycles. However, the battery capacity does not strictly follow a specific changing trend in actual application. It is also influenced by many other factors. For instance, in some cycles, the capacity may exhibit an increasing trend due to transient and recovery effects inside the battery [19]. Additionally, variations in the operating conditions and load profiles of the battery can cause fluctuations in the degradation curve of SOH. Therefore, identifying HFs that accurately reflect the battery's health state is one key for SOH diagnosis and prognosis. In actual application, batteries are typically replaced when their capacity drops below 80%. Hence, in the experiments, the aging test is terminated when the battery capacity drops below 80%. The experimental data shows that the battery capacity drops to $(580)/750 \times 100\% = 77.3\%$ after 250 cycles.

B. Health feature extraction

This section analyzes some basic aging mechanisms of LIBs and the accessibility and effectiveness of HFs to determine their use for SOH estimation. To meet actual application requirements, data should be easily obtainable and accessible. In actual application, the discharge process is often unpredictable due to various factors such as load variations and environmental conditions. However, compared to the discharge process, the charging process is typically standardized and more controllable. Based on this, HFs of the battery are obtained from the voltage curves during the typical charging process to acquire effective HFs. The extraction of these health features can serve as a key step in developing battery capacity models, contributing to improving the performance and reliability of battery systems.

Fig. 3 shows the changing trend of the terminal voltage at 1 C-rate with the aging cycles. From the first to the 200-th aging cycle, the overall changing trend of the terminal voltage shifts from the right to the left. Specifically, as the battery ages, the charging time from 4.0 V to 4.2 V decreases. In addition, the rise rate of the terminal voltage increases with an increase in the number of aging cycles, and the constant current charging time decreases. These phenomenons are attributed to various factors, including aging mechanisms, over-potential increases, and particle cracking, etc. The aging mechanism refers to changes in the internal structure of the battery, leading to capacity decay and increased internal resistance. Over-potential increases due to slower electrochemical reaction rates during battery charge and discharge processes, requiring higher over-potentials, thereby reducing battery efficiency. Partical cracking refers to stress accumulation caused by the expansion and contraction of electrode materials during cycling, which may lead to delamination and cracking of electrode materials, affecting battery performance and lifespan.

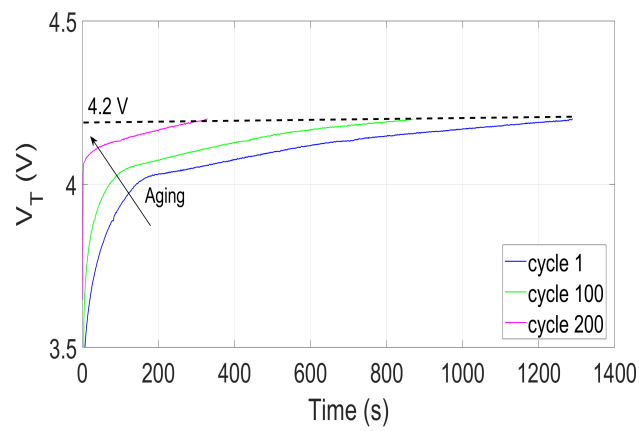

Fig. 3: Change of terminal voltage in battery aging experiment

Based on the changing trend of the terminal voltage with aging cycles, the HFs are described as follows:

(1) CHT: It is defined as the time duration for the battery charging from 4.0 V to 4.2 V. Fig. 3 shows that, as a LIB ages, CHT will increase. Besides, the time duration of the battery charging from 4.0 V to 4.2 V is usually short at typical 1 C-rate charging, which make it efficient to extract health feature.

(2) $Ratio_{\mathrm{cccv}}$: It is defined as the ratio between the capacity of constant-current charge from 4.0 V to 4.2 V and capacity of

the constant-voltage charge at 4.2 V before the current drops to the cut-off current. The charging process can be divided into two stages, i.e. the first stage is constant current charging, and the second stage is constant voltage charging. For the fresh battery, the capacity and energy stored in the first stage is much larger than the second stage. As battery ages, the capacity and energy stored in the first stage (second stage) decreases (increases). This is mainly because the over-potentials increase as the battery ages, which make the charging terminal voltage reaches 4.2 V faster, and the charging mode switches to the second stage earlier. Therefore, $Ratio_{\mathrm{cccv}}$ can be selected as one HF candidate.

(3) R_{ohm}: The resistance R_{ohm} accounts for the Ohmic polarization and it acts much faster than solid-phase and liquid-phase diffusion processes. The resistance R_{ohm} increases as the LIB ages and it reflects the battery health state. The resistance R_{ohm} is calculated as the ratio of ΔU and I:

$$R_{\mathrm{ohm}} = \frac{\Delta U}{I} \tag{2}$$

where ΔU is the instant variation of terminal voltage at the moment when a discharging current I is applied to a fully charged battery after a period of rest.

Next, the relationship between battery aging and these HFs are illustrated with experimental data. Fig. 4 shows the change of the capacity Q_{all} with the aging cycles. This indicates that the battery capacity gradually decreases as the number of aging cycles increases.

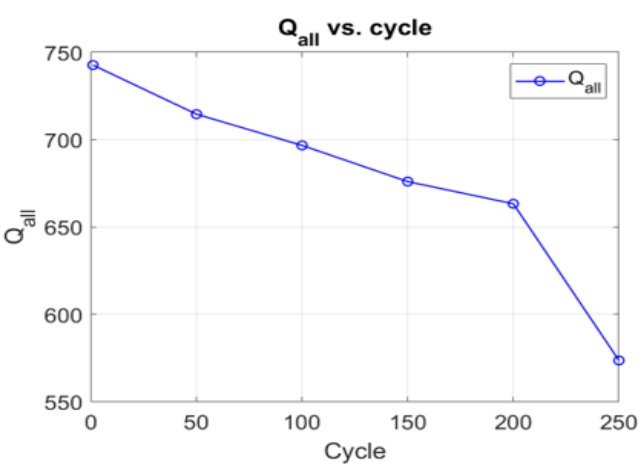

Fig. 4: Q_{all} against aging cycles

Fig. 5 shows the change of CHT with the aging cycles. This phenomena is caused by the fact that the polarization over-potential continues to increase as a LIB ages. As a result, the time required for the battery to reach the constant voltage at 4.2 V charging condition decreases.

Fig. 6 shows the change of $Ratio_{cccv}$ with the aging cycles. The reason is that, with the aging of the battery, the lithium-ion diffusion process becomes more difficult. Therefore, the time required to eliminate polarization in constant voltage charging process increases. As a result, the amount of Li^+ that are absorbed in constant current process decreases while that in constant voltage charging process increases.

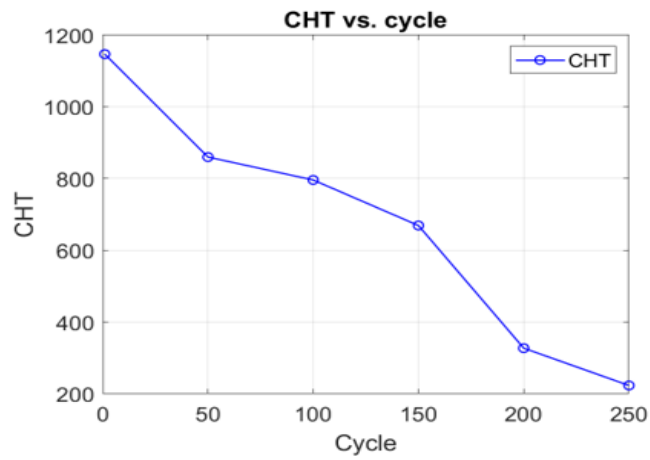

Fig. 5: CHT against aging cycles

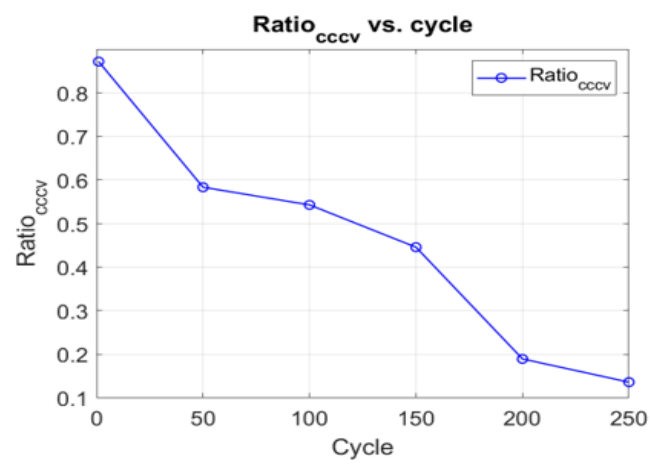

Fig. 6: $Ratio_{\mathrm{cycle}}$ against aging cycles

Fig. 7 shows the increasing variation of R_{ohm} with the aging cycles. The reasoning behind this phenomena is that, the aging of the battery leads to the increase in the amount of cracking particles, solid electrolyte interphase (SEI) film formulation, lithium-ion dendrite, and aging of electrode contact surfaces. All of these factors result in the increase of R_{ohm}.

The changes of HFs with the aging cycles are further summarized in Table II. It is worth noting that these HFs are extracted from online data (V_{T} and I) of LIBs in aging experiments, which indicates that they are easy-to-access from in-situ battery application.

III. Model Establishment and Validation

A. Correlation of Q_{all} and HFs

According to the changing trend of CHT, $Ratio_{\mathrm{cccv}}$, and R_{ohm} with HFs, the linear fitting method is used to obtain the relationship between HFs and capacity Q_{all}. Figs. 4, 5, 6, and 7 show that Q_{all} is positively correlated with

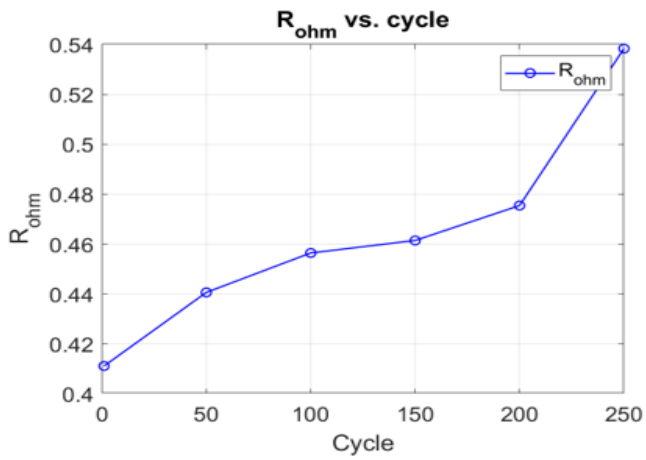

Fig. 7: R_{ohm} against aging cycles

TABLE II: Change of HFs with the aging cycles

Cycle	CHT (s)	$Ratio_{\text{cccv}}$	R_{ohm} (mohm)
1	1147	0.8812	0.4111
50	860	0.5833	0.4406
100	796	0.5428	0.4564
150	669	0.4457	0.4614
200	327	0.1893	0.4754
250	223	0.136	0.5383

CHT and $Ratio_{cccv}$, and negatively correlated with R_{ohm}, respectively. The correlations between Q_{all} and HFs indicate the effectiveness of the selected HFs. However, the degree of each correlation is different. In order to comprehensively incorporate information of different HFs for SOH estimation model development, the coefficient of determination R^2 in Eq. (3) is used to evaluate the correlation between Q_{all} and HFs. The closer of R^2 to 1, the better the correlation is [20].

$$R^2 = 1 - \frac{\sum_i (y_i - f_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{3}$$

where i represents the data point index, y_i represents the value of the ith data point, $\bar{y}$ represents the mean value of the total data point, f_i represents the value of the ith data point calculated by the fitted linear model. Note that data show that even though both the changing trends of HFs and Q_{all} with cycle number are nonlinear, especially at later battery aging stages, the changing trends of HFs with Q_{all} are relatively linear. Therefore, a linear fitting method is used to obtain the relationship between HFs and capacity Q_{all}.

B. Model establishment

Using linear fitting method, the variations of R^2 of Q_{all} with the HFs are calculated. R^2 between Q_{all} and CHT, $Ratio_{\text{cccv}}$, and R_{ohm} are 0.8332, 0.7912, and 0.99 respectively.

The result shows that Q_{all} has good correlation with HFs of CHT and R_{ohm}, but has relative low correlation with $Ratio_{\text{cccv}}$. In order to comprehensively and reasonably take advantage of the battery aging information in HFs, a weighted linear prediction model for Q_{all} is expressed as:

$$\begin{aligned} Q_{\text{all}} = & \omega_{Q_{\text{all}}-\text{CHT}} \cdot f_{Q_{\text{all}}-\text{CHT}} + \\ & \omega_{Q_{\text{all}}-Ratio_{\text{cccv}}} \cdot f_{Q_{\text{all}}-Ratio_{\text{cccv}}} + \\ & \omega_{Q_{\text{all}}-R_{\text{ohm}}} \cdot f_{Q_{\text{all}}-R_{\text{ohm}}} \end{aligned} \tag{4}$$

where $f_{Q_{\text{all}}-\text{CHT}}$, $f_{Q_{\text{all}}-Ratio_{\text{cccv}}}$, and $f_{Q_{\text{all}}-R_{\text{ohm}}}$ are fitting function of Q_{n} and CHT, $Ratio_{\text{cccv}}$, and R_{ohm}, respectively. Here, $\omega_{Q_{\text{all}}-\text{CHT}}$, $\omega_{Q_{\text{all}}-Ratio_{\text{cccv}}}$, and $\omega_{Q_{\text{all}}-R_{\text{ohm}}}$ are:

$$\omega_{Q_{\text{all}}-\text{CHT}} = \frac{R^2_{Q_{\text{all}}-\text{CHT}}}{R^2_{Q_{\text{all}}-\text{CHT}} + R^2_{Q_{\text{all}}-Ratio_{\text{cccv}}} + R^2_{Q_{\text{all}}-R_{\text{ohm}}}}$$

$$\omega_{Q_{\text{all}}-Ratio_{\text{cccv}}} = \frac{R^2_{Q_{\text{all}}-Ratio_{\text{cccv}}}}{R^2_{Q_{\text{all}}-\text{CHT}} + R^2_{Q_{\text{all}}-Ratio_{\text{cccv}}} + R^2_{Q_{\text{all}}-R_{\text{ohm}}}}$$

$$\omega_{Q_{\text{all}}-R_{\text{ohm}}} = \frac{R^2_{Q_{\text{all}}-R_{\text{ohm}}}}{R^2_{Q_{\text{all}}-\text{CHT}} + R^2_{Q_{\text{all}}-Ratio_{\text{cccv}}} + R^2_{Q_{\text{all}}-R_{\text{ohm}}}}$$

where $R^2_{Q_{\text{all}}-\text{CHT}}$, $R^2_{Q_{\text{all}}-Ratio_{\text{cccv}}}$, and $R^2_{Q_{\text{all}}-R_{\text{ohm}}}$ are R^2 values between Q_{all} and CHT, $Ratio_{cccv}$, and R_{ohm}, respectively. The weights between capacity Q_{all} and HFs of CHT, $Ratio_{cccv}$, and R_{ohm} are 0.3187, 0.3026, and 0.3787, respectively. Note that different HFs have different degrees of relevance and relationships with battery aging state. To reasonably reflect the degree of relationship between HF and battery aging state, weighting factors $\omega_{Q_{\text{all}}-\text{CHT}}$, $\omega_{Q_{\text{all}}-Ratio_{\text{cccv}}}$, and $\omega_{Q_{\text{all}}-R_{\text{ohm}}}$ corresponding to each fitting function are introduced as above.

C. Model validation

This section presents the validation of the developed weighted model for SOH estimation with the experimental aging data on a different battery sample. The variations of Q_{all} and HFs of the new battery sample with the aging cycles are summarized in Table III. These data are used for the validation of the developed weighted SOH estimation model. It is worth noting that these HFs are extracted from online data (V_{T} and I) of the LIB in aging experiments.

TABLE III: Changes of Q_{all} and HFs of the new battery sample with the aging cycles

Cycle	CHT (s)	$Ratio_{\text{cccv}}$	R_{ohm}(mohm)	Q_{all} (mAh)
1	1560	0.914	0.315	731
50	1500	0.83	0.332	724
100	1350	0.70	0.360	711
150	1170	0.57	0.378	691
200	840	0.41	0.389	638

With the weighted SOH estimation model expressed in Eq. (4) and the HFs in Table III, the SOH estimation results of the new battery sample are summarized in Table IV.

Fig. 8 shows the SOH estimation result of the new battery sample with the developed SOH estimation model. From the first aging cycle to the 200-th cycle, the maximum error between the experimental capacity and estimated capacity is $(691 - 667)/691 * 100\% = 3.47\%$, which is at the 150-th

TABLE IV: SOH estimation results of the new battery sample based on the weighted SOH estimation model

Cycle	$Q_{\text{exp.}}$ (mAh)	$Q_{\text{est.}}$ (mAh)	Q_{error} (mAh)	error (%)
50	724	716	8	1.10
100	711	690	21	2.95
150	691	667	24	3.47
200	638	640	2	0.31

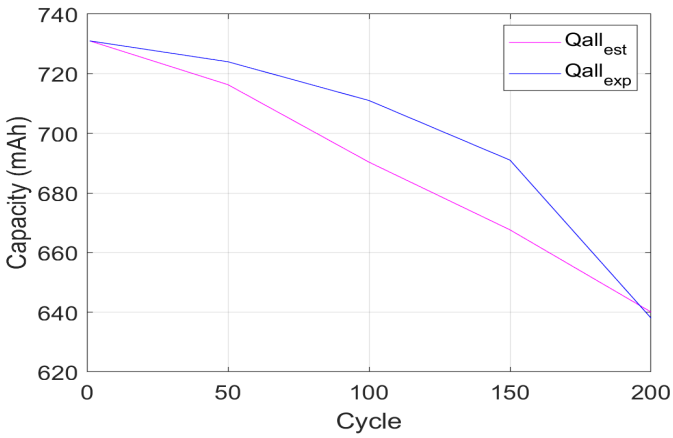

Fig. 8: Comparison of experimental capacity and the estimated capacity by the weighted SOH estimation model

aging cycle. The developed weighted SOH estimation model shows overall performance of SOH estimation.

Note that the developed weighted SOH estimation model needs to be adjusted by an adjusting factor f_{adj} expressed in Eq. (5). With the extracted HFs of a new battery sample, f_{adj} needs to be adjusted to make the estimated capacity matches the experimental capacity at the first aging cycle. The f_{adj} is then used for SOH estimation in following aging cycles.

$$\begin{aligned} Q_{\text{all}} = f_{\text{adj}} * (&\omega_{Q_{\text{all}}-\text{CHT}} \cdot f_{Q_{\text{all}}-\text{ CHT}} + \\ &\omega_{Q_{\text{all}}-Ratio_{\text{cccv}}} \cdot f_{Q_{\text{all}}-Ratio_{\text{cccv}}} + \\ &\omega_{Q_{\text{all}}-R_{\text{ohm}}} \cdot f_{Q_{\text{all}}-R_{\text{ohm}}}) \end{aligned} \tag{5}$$

IV. Conclusion

LIBs usually work in varying, unpredictable, and complicated operating conditions for different application. It is therefore challenging to develop a reliable and practical SOH estimation method for in-situ cells based on the easy-to-access data in actual application. Based on the changing trends of the terminal voltage and the aging mechanisms behind them, our research extracts HFs from battery aging data, and develops a weighted SOH estimation model. The HFs reflect battery aging phenomenon and can be easily extracted from the in-field battery data. The weighted SOH estimation model comprehensively incorporates battery aging information by weighting the HFs. The validation results on new battery samples show the effectiveness of the proposed weighted SOH estimation model. Our future work will focus on more battery aging experiments of more battery samples with more work conditions, and more feature extraction to develop a more generative SOH estimation model. Moreover, our future work will also focuses on the battery packs that are often used to meet the voltage and power requirements for industrial application.

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